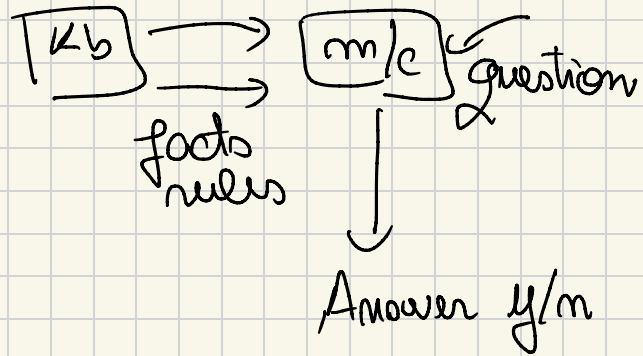


Notiuni introductive

John owns the book.
person owns obj.



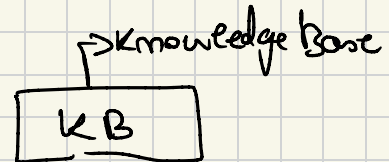
PROLOG vs. C++ / Java

- data structure
- objects

Facts

- likes (john, mary)
relationship obj
- gives (john, book, mary)
- play (jane, mary, football)

ex: owns (jane, gold)



comparat cu owns (mary, book)
No

Instructiuni

- ? - întrebare ⇒ poate să fie declarat înainte
da/no (nu diferit de false)
ordinea parametrilor contează

dacă am declarat cu un parametru ⇒ eroare dacă pun mai mulți parametri

- . ⇒ ;

Variable

- în compiler : likes(john, X).
↳ variabilă se declară cu majusculă
X = flowers. sau X = obj1;
X = obj2.

- elev a 12-a, BAC → Mihai Nemes
o dată la 2 sapt.

mărcuți, vimeu

AND / OR

? - likes(john, mary) $\xrightarrow{\text{AND}}$ ① likes(mary, john).

T F

AND - ,

T, F → F T, T → T
F, T → F F, F → F

OR - ;

T, F → T T, T → T
F, T → T F, F → F

! dacă prima este falsă, nu mai este verif. a doua

IF (⇒) :- ex: isolder(X, Y) :- female(X), parents(X, M, F), parents(Y, M, F)

likes(mary, wime)

likes(john, X) :- likes(X, wime)

? - likes(john, mary) ⇒ True

Caractere, constante, variabile

→ caractere (A B C ... Z, a b c ... z, 0 1 ... 9,
* - \ / ~ ^ < > ...)

NU: Blank Space, Newline, Beeping Sound

→ constante

• numere (-17, 18, 1.01e2, 67, 1.01e-2)

• atoms (string) → literă + cifră

NU
?
:-

signs
ex: * + -
+ \ /

ex: aab1
laab ✗
bev ✗
cbev

matt - p ✗
① 2018a ① ✓

'matt - Boston

→ variable (beg. with uppercase / underscore)

ex: Answer
_rez

Variable anonime

hates (_ , oswald) - cineva îl urăște pe oswald

exemple/aplicații:

- $2 + ((3/2) * 3) \Rightarrow + (2, * ((3/2), 3))$
- $2 + 3 * 4 \Rightarrow + (2, 12)$
14 20
- 'student' = student $\Rightarrow$ true

Operatori aritmetici

$x == y$ equal
 $x != y$ not equal
 $x < y$, $x > y$
 $x // y$ împărțire exact
 $x \bmod y$

Găsim elem în listă

[1, 2, 3, 4, 5] (3)

member(x, [x | _]).

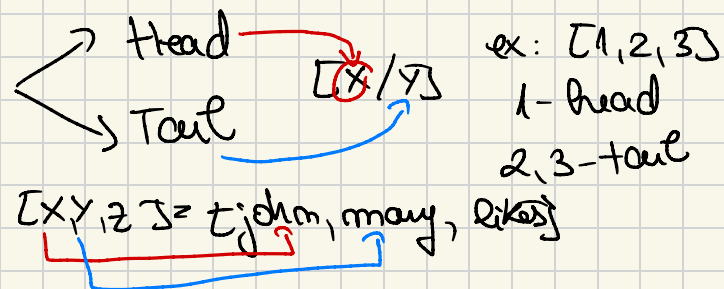
member(x, [_ | T]) :- member(x, T).

0	1	2
H=1 T=[2,3,4] 3=1? X ① return	H=2 T=[3,4] 3=2? X ② return	H=3 T=[4] 3=3? V true return
T	T	

List

[] empty list

[1, 2, 3] [1, [2, 3]]



lungimea unei liste

$[1, 2, 3, 4] \Rightarrow 4$.

$\text{size}([], 0)$.

$\text{size}([_H|_T], N) :- \text{size}(T, N1), N \text{ is } N1 + 1$

$H=1$	$H=2$	$H=3$	$H=4$	first cond
$T=[2,3,4]$	$T=[3,4]$	$T=[4]$	$T=[]$	0
$N=N1+1$	$N=N1+1$	$N=N1+1$	$N=N1+1$	

$\begin{matrix} \nearrow & \nearrow & \nearrow & \nearrow \\ =4 & =3 & =2 & =1 \end{matrix}$

Suma elem unei liste

$[1, 2, 3]$

$\text{sumlist}([], 0)$.

$\text{sumlist}([_H|_T], N) :- \text{sumlist}(T, N1), N \text{ is } N1 + H$

Căutare recursivă

$[adam, john, hanny, jama]$ $\nwarrow \neq$

$\text{member}(x, [_x|_T])$.

$\downarrow \quad \downarrow$
hanny adam

$\text{member}(x, [_y|_T]) :- x = y$.

Check a list is sorted

$[1, 4, 7, 9]$ compar elem. cu elem.

is - sorted $([])$. empty list

is - sorted $([_x|_T])$. 1 elem.

Append / concatenated

$\text{append}([], L2, L2)$.

$\text{append}([_H|_T], L2, [_H|_L3]) :- \text{append}(T, L2, L3)$